

DATA 4380

Celebrity Face Identification



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Introduction

- In this project, we built a complete deep-learning pipeline for **celebrity face identification** using 18 celebrity classes.
- We developed and evaluated multiple models, starting from a simple CNN baseline to advanced **transfer learning models** including VGG16 and DenseNet121.

Goals

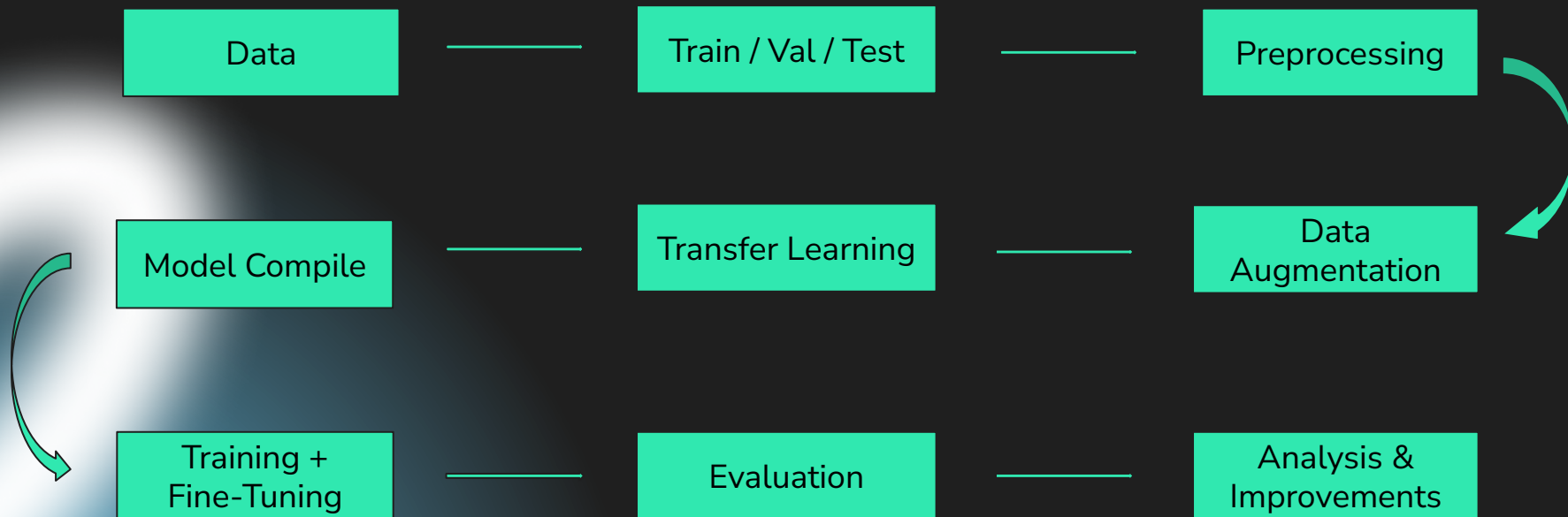
- The objective of the project is to evaluate how different model complexities, feature extraction techniques, and fine-tuning strategies influence classification accuracy.
- Through systematic experimentation, the team aimed to build a model capable of reliably identifying celebrity faces despite variations in lighting, pose, background, and overall image quality.



Dataset : Celebrity Face Image

- 18 Different Celebrities
- 1800 Images
- Each Celebrity - 100 Images
 - Scarlett Johansson (200)
- Training Set: 1260 Images
- Test Set: 270 Images
- Validations Set: 270 Images

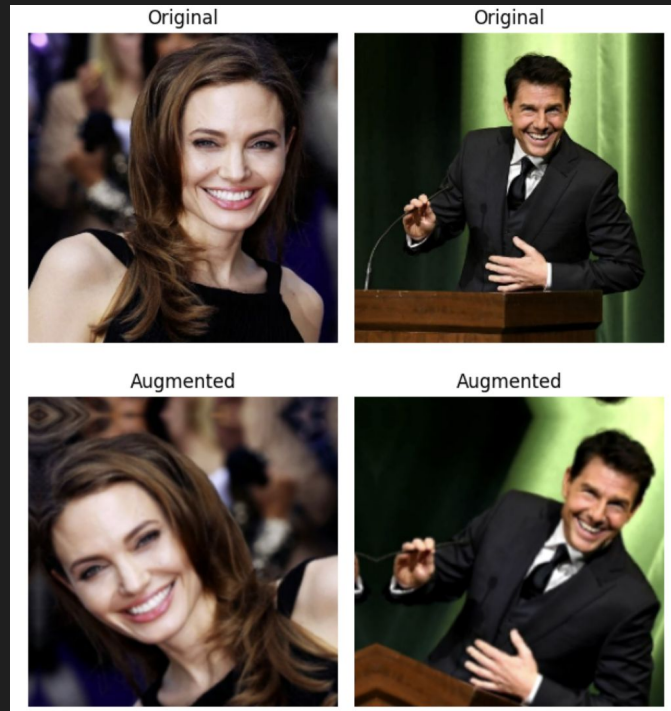
Pipeline



Data Preprocessing

Things we did:

- Rescaled Pixel Values
- Data Augmentation
 - Rotation
 - Zoom
 - Flips
 - Contrast / Brightness
- 70/15/15 Split



Baseline Model

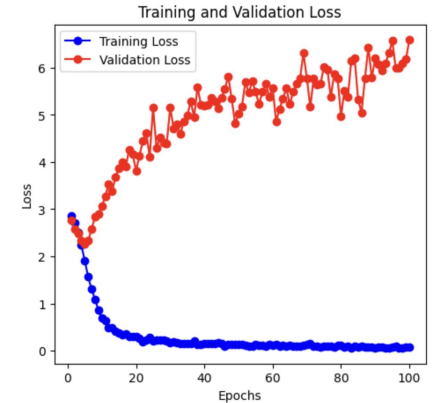
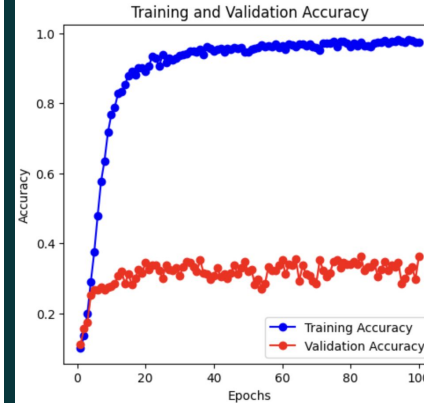
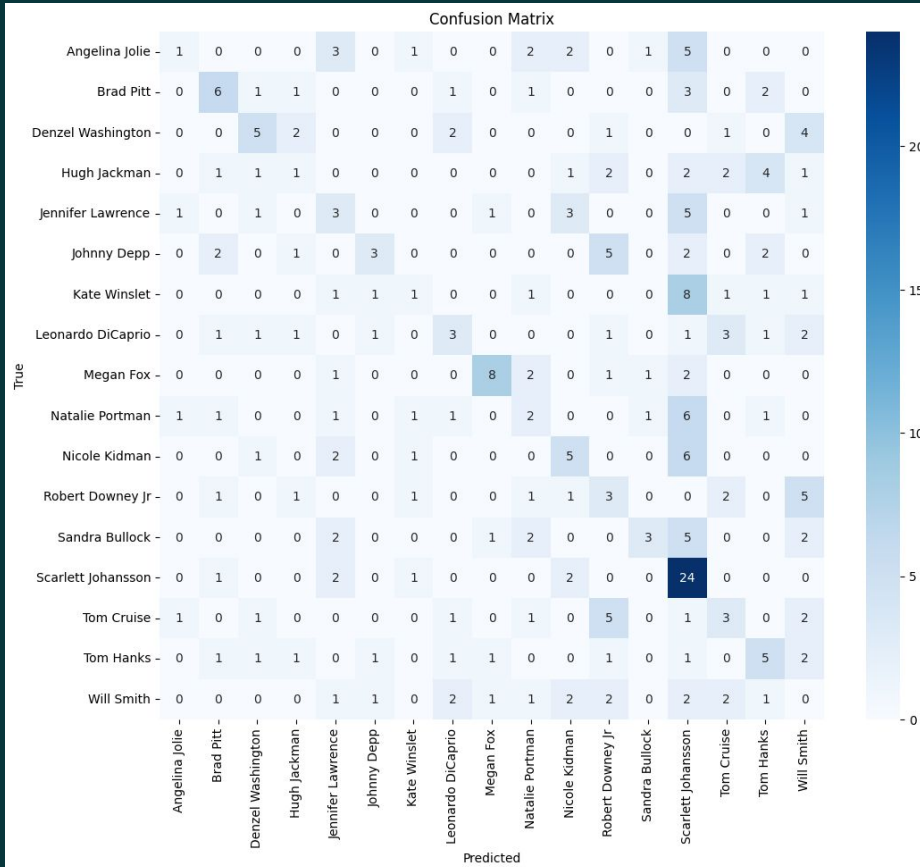
CNN Architecture (100 Epochs)

- Input Layer
- Block 1
 - 32 Filters / Relu / Padding
- Block 2
 - 64 Filters / Relu / Padding
- Block 3
 - 128 Filter / Relu / Padding
- Flatten Layer
- Dense Layer
- Dropout (0.5)
- Outer Layer
 - Classes: 18
 - Softmax

Classification Report:

	precision	recall	f1-score	support
Angelina Jolie	0.29	0.27	0.28	15
Brad Pitt	0.31	0.27	0.29	15
Denzel Washington	0.29	0.53	0.37	15
Hugh Jackman	0.12	0.07	0.09	15
Jennifer Lawrence	0.46	0.40	0.43	15
Johnny Depp	0.31	0.33	0.32	15
Kate Winslet	0.27	0.27	0.27	15
Leonardo DiCaprio	0.20	0.27	0.23	15
Megan Fox	0.64	0.60	0.62	15
Natalie Portman	0.31	0.27	0.29	15
Nicole Kidman	0.44	0.53	0.48	15
Robert Downey Jr	0.20	0.20	0.20	15
Sandra Bullock	0.50	0.33	0.40	15
Scarlett Johansson	0.66	0.83	0.74	30
Tom Cruise	0.24	0.27	0.25	15
Tom Hanks	0.50	0.33	0.40	15
Will Smith	0.25	0.13	0.17	15
accuracy			0.37	270
macro avg	0.35	0.35	0.34	270
weighted avg	0.37	0.37	0.36	270

Baseline Model



Transfer Learning Model: VGG16

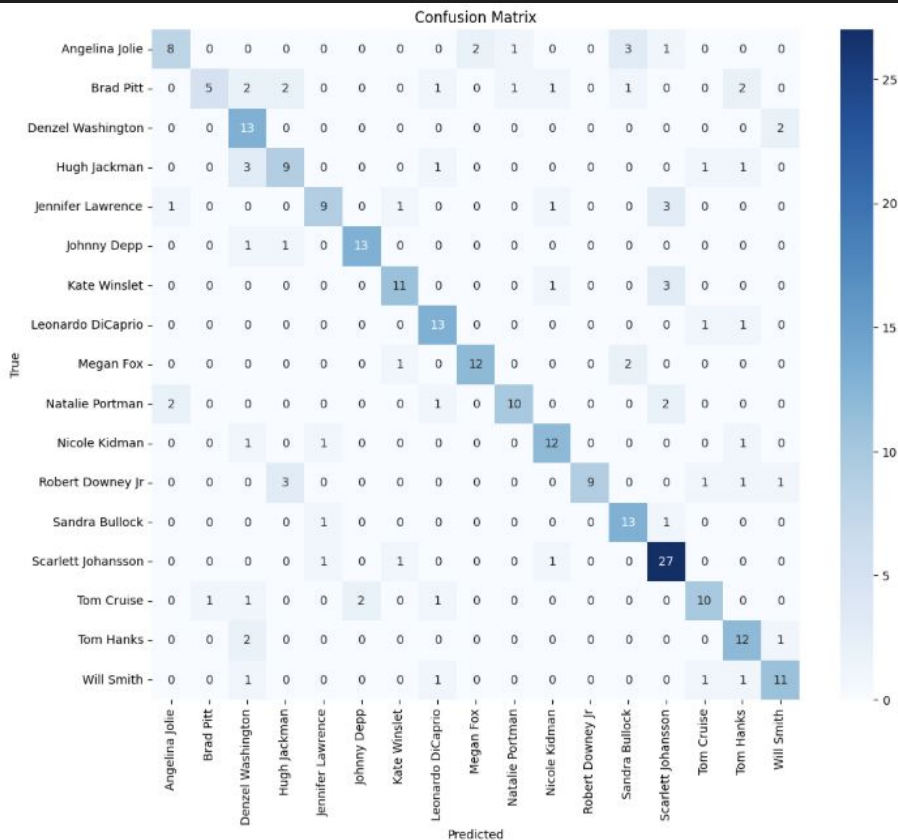
CNN Architecture (50 Epochs)

- Input Layer
- VGG16
 - 13 ConV Layers + 3 Max Pooling Layers
- Custom CNN
 - GlobalAveragePooling2D
 - Dense(512, relu)
 - Dropout
- Outer Layer
 - Classes: 18
 - Softmax
- Unfreeze last VGG16 Block

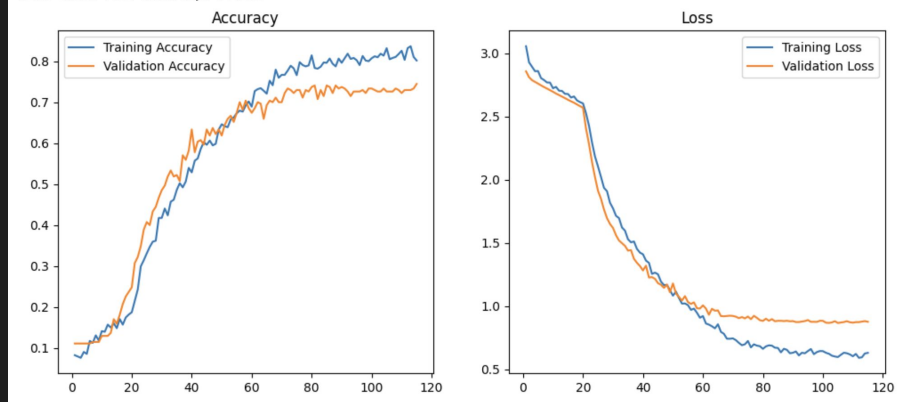
CLASSIFICATION REPORT:

	precision	recall	f1-score	support
Angelina Jolie	0.73	0.53	0.62	15
Brad Pitt	0.83	0.33	0.48	15
Denzel Washington	0.54	0.87	0.67	15
Hugh Jackman	0.60	0.60	0.60	15
Jennifer Lawrence	0.75	0.60	0.67	15
Johnny Depp	0.87	0.87	0.87	15
Kate Winslet	0.79	0.73	0.76	15
Leonardo DiCaprio	0.72	0.87	0.79	15
Megan Fox	0.86	0.80	0.83	15
Natalie Portman	0.83	0.67	0.74	15
Nicole Kidman	0.75	0.80	0.77	15
Robert Downey Jr	1.00	0.60	0.75	15
Sandra Bullock	0.68	0.87	0.76	15
Scarlett Johansson	0.73	0.90	0.81	30
Tom Cruise	0.71	0.67	0.69	15
Tom Hanks	0.63	0.80	0.71	15
Will Smith	0.73	0.73	0.73	15
accuracy			0.73	270
macro avg	0.75	0.72	0.72	270
weighted avg	0.75	0.73	0.72	270

Transfer Learning Model: VGG16



Fine-tuned Test Accuracy: 0.7296



Transfer Learning Model: DesNet121

CNN Architecture

- Input Layer (256,256,3) & Scaled
- DenseNet121
- Custom CNN
 - GlobalAveragePooling2D
 - Dense(512, relu)
 - Dropout
- Outer Layer
 - Classes: 18

Fine Tuning:

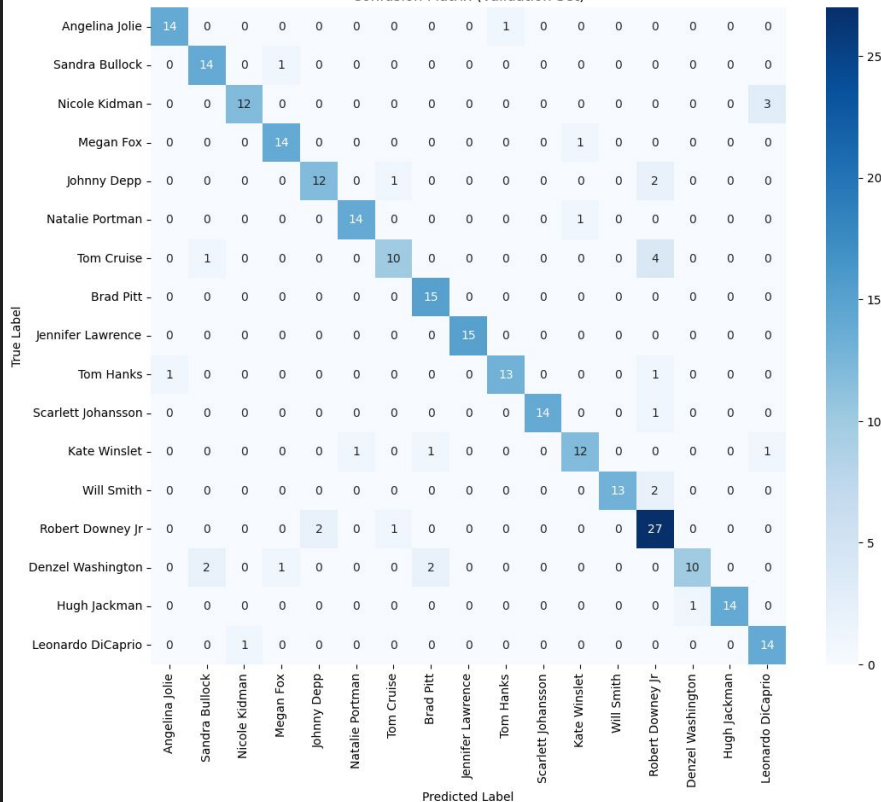
- Unfreeze Base Model
- Recompile Model
- Trained Additional 60 Epochs

Classification Report (Test Set):

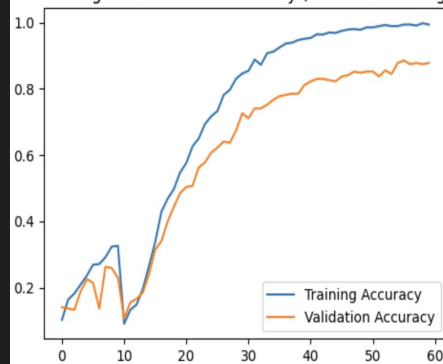
	precision	recall	f1-score	support
Angelina Jolie	0.91	0.67	0.77	15
Sandra Bullock	0.65	0.73	0.69	15
Nicole Kidman	0.93	0.87	0.90	15
Megan Fox	0.82	0.93	0.88	15
Johnny Depp	0.71	0.80	0.75	15
Natalie Portman	0.88	0.93	0.90	15
Tom Cruise	0.79	0.73	0.76	15
Brad Pitt	0.80	0.80	0.80	15
Jennifer Lawrence	0.92	0.80	0.86	15
Tom Hanks	0.86	0.80	0.83	15
Scarlett Johansson	1.00	0.73	0.85	15
Kate Winslet	0.80	0.80	0.80	15
Will Smith	0.91	0.67	0.77	15
Robert Downey Jr	0.69	0.97	0.81	30
Denzel Washington	0.92	0.73	0.81	15
Hugh Jackman	1.00	0.93	0.97	15
Leonardo DiCaprio	0.88	1.00	0.94	15
accuracy			0.83	270
macro avg	0.85	0.82	0.83	270
weighted avg	0.84	0.83	0.83	270

Transfer Learning Model: DenseNet121

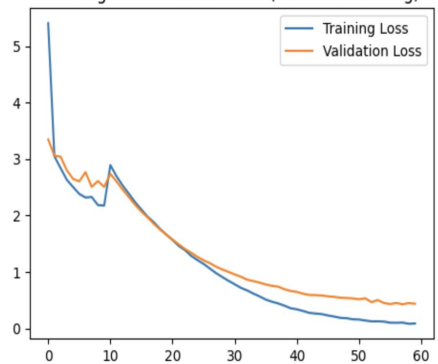
Confusion Matrix (Validation Set)



Training and Validation Accuracy (Transfer Learning)

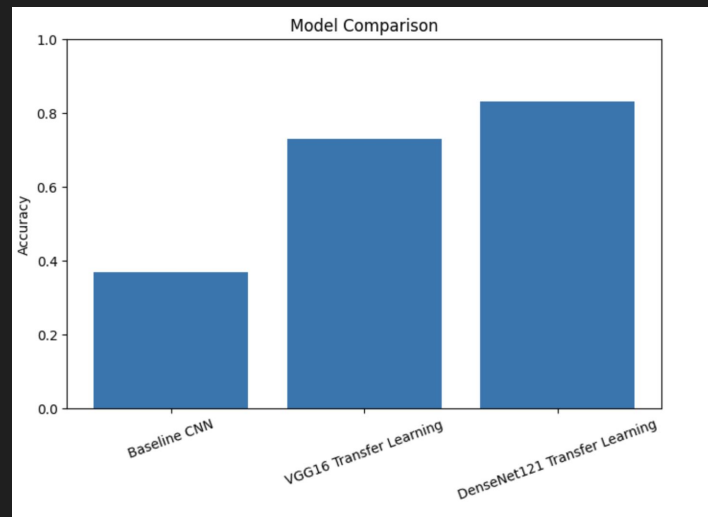


Training and Validation Loss (Transfer Learning)



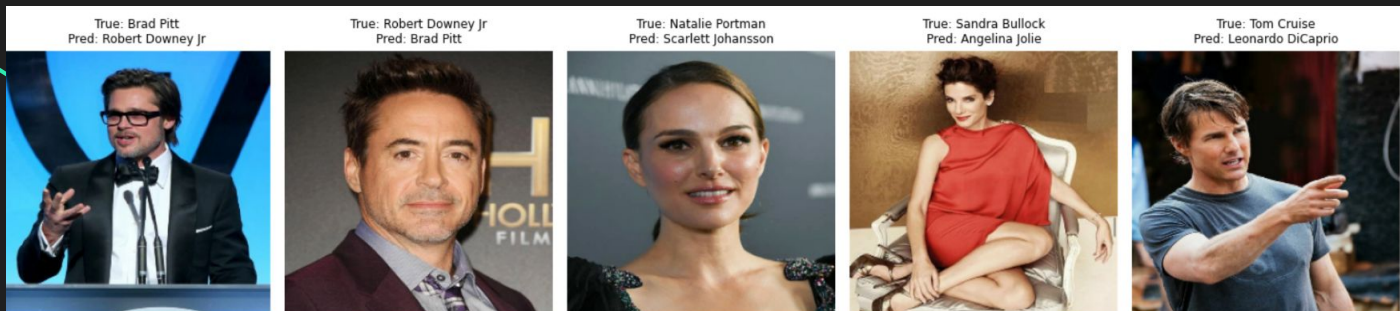
Conclusion

- **Baseline Model:**
 - Accuracy: 37%
 - Overfitting
 - Misclassifications
- **VGG16 Transfer Learning Model**
 - Accuracy: 73%
 - Mild Overfitting
 - Less Misclassifications
- **DenseNet121 Transfer Learning Model**
 - Test Accuracy: 83%
 - Mild Overfitting
 - Prediction True Labels at higher level



Challenges

- Limited Dataset Size
- Inconsistent Image Quality
- High Intra-Class Variation
- Inter-Class Similarity
- Overfitting in Baseline CNN Models
- Pose / Angle Sensitivity
- Transfer Learning Complexity
- Nested Folders in Directory



Future Works

- **Collect a Larger and More Balanced Dataset**
Increasing the number of images per celebrity would significantly reduce overfitting and improve generalization.
- **Implement Face Alignment Techniques**
Using landmark detection to standardize angle, orientation, and cropping may improve performance on angled/profile images.
- **Experiment With Additional Transfer Learning Models**
Testing architectures such as EfficientNet, MobileNetV3, and Vision Transformers for higher accuracy and lower computational cost.
- **Hyperparameter Tuning**
Using systematic search methods (e.g., KerasTuner, Optuna) to optimize learning rate, batch size, and dropout.
- **Model Fine-Tuning**
Unfreezing deeper layers of VGG16 or DenseNet121 for improved feature extraction on celebrity-specific patterns.
- **Use Face Embedding Approaches**
Exploring methods like FaceNet or DeepFace + SVM/XGBoost classifiers for better performance on small datasets.

Citations

Alblushi, A. (2021). Face Recognition Based on Artificial Neural Network: A Review. *Artificial Intelligence & Robotics Development Journal*, 1(Issue 2), 116-131.
<https://doi.org/10.52098/airdj.202125>

Kasim, N. A. B. M., Rahman, N. H. B. A., Ibrahim, Z., & Mangshor, N. A. (2018). Celebrity face recognition using deep learning. *Indonesian Journal of Electrical Engineering and Computer Science*, 12(2), 476-481. <https://d1wqtxts1xzle7.cloudfront.net/103515497/9380-libre.pdf?>

Le, Thai Hoang, Applying Artificial Neural Networks for Face Recognition, *Advances in Artificial Neural Systems*, 2011, 673016, 16 pages, 2011. <https://doi.org/10.1155/2011/673016>

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